CoFrance

User Manual

French vACC

Version v1.0.7

Contents

1 Disclaimer

Even though the development of this plugin was inspired by real ATM systems such as those from Thales Group, the plugin is absolutely NOT affiliated with or endorsed by said company. The plugin and all associated files shall be used solely for the purpose of simulation on the VATSIM network.

2 Setup

2.1 Installation

- Create a CoFrance folder in `Plugins`, this is where all the setup will take place
- Copy the `CoFranceLoader.dll` in the CoFrance folder.
- Ensure Control Panel > Clock & Region > Change date, time or number format > Format: French or English (UK)
- Unload both the legacy version of CoFrance and PINEDE: open `.prf` file using a text editor and remove lines `PLUGINx:CoFrance` & `PLUGINx:PINEDE`
- Start EuroScope and load the plugin, the updater will download required files.
- Allow `CoFrance` to draw on all radar types (including vSMR).
- Allow `RDF plugin` to draw on `CoFrance radar display`

You are done with the installation process! The plugin should now be running with no errors.

2.2 First Usage

This section lists recommendations for controllers that aren't familiar with this environment, or those switching from the legacy version of CoFrance.

The environment brought by this plugin is most likely different from the one you have been previously using, and various features will be confusing. Therefore it is sensible to try it for the first time during <u>low traffic conditions</u>.

We draw your attention specifically to some User Settings marked with an $*$ in the next section.

2.3 User Settings

To open the user settings window, open the `Radar Menu` using MOD+🖱 (if first open, `MOD` key is `CTRL`) on the radar screen background and head down to `Setup`.
You can create multiple settings profiles that will be linked to the `ASR` file and auto reloaded when switching to said `ASR`.
Profiles are saved inside `CoFrance_Preference.toml` file.

∗ **CPDLC API key:** Your CPDLC API token [1]
Auto Focus Entry Field: When opening popups, keyboard input field will be directly focused.
Auto Mouse Positionning: Mouse is moved to specific buttons automatically.
Use RECAT-EU Categories: Display RECAT-EU Categories next to ATYP.
∗ **Filter Unconcerned Only:** Prevent filtering of Assumed or Notified flights.

∗ **Select Modifier (MOD) Key:** Let you select the key used to trigger numerous actions *(CTRL by default)*.
Modifier (MOD) Extended Tag Scrolling: Modifier key needed to display extended tag.
Hold Indicator Always Visible: Hold indicator will be visible in non-detailled tag.
Unconcerned Tag Start Minimised: Unconcerned flights tag's will be minimised by default.
Seen / Not Seen Enabled: Enable Seen / Not Seen logic.
Auto Flight Leg: Automatically display flight leg when a not seen tag is detailed.
Sound Notification Enabled: Enable sound notification for SHRQ/POINT/CPDLC Requests/...
STCA Alert Sound: Enable STCA Alert Sound independently from other sound notification.
Uncontrolled Intrusion Alert Enabled: Enable Intrusion Alert for uncontrolled aircrafts.
Tag Font Size Offset: Allows to change Tag font size from -2 to +2.
List Font Size Offset: Allows to change Lists font size from -2 to +2.
Enable STCA: Enable STCA calculations & alert.
Enable TCT: Enable TCT calculations & alert.
∗ **Use Scroll Wheel for Speed Vector:** Allow scrolling entire screen to modify speed vector.
(Note: this blocks EuroScope from zooming the radar, if you wish to enable this option, F11 and F12 should be used for zooming).
Display COPX in Tagged Layout: Displays COPX when tag is not detailed.
Use ACC Custom Intention Codes: Display custom code when applicable in place of ADES (not recommended).
Restricted Areas Enabled: Enable parsing of active restricted areas from API.
Auto Window Hide: Hides all CoFrance windows on non-CoFrance ASRs.

2.4 Customization

All files except `CoFrance_Preference.toml`, should not be tempered with as they are already pre-configured and automatically updated on plugin start-up.

[1] To paste your API key, open Plugins/CoFrance/config/CoFrance_Preference.toml and modify cpdlc_api_key value.

2.5 StreamDeck Setup

Multiple CoFrance actions can be triggered through a StreamDeck device, they are recognizable by the statement: `This action is available through StreamDeck integration.`
First, install `CoFrance StreamDeck Integration` plugin.
Then, install the provided profile for a quick setup, otherwise, find all the available actions under the `CoFrance` tab in the StreamDeck UI.

`Speed Vector` actions require the user to select the length in the setting of the button and then select the corresponding icon from the provided icon pack.

3 Usage

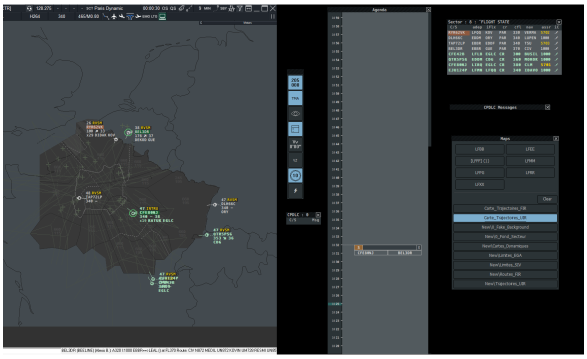
3.1 General Introduction

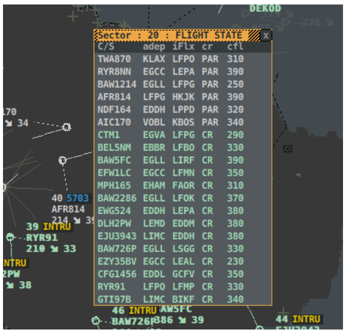
Graphical user interface is coherent across the entire plugin visual indications:

Blue and underlined always depicts a selected item or `true` state.

A crosshair cursor indicates that a tool is currently enabled, to leave a tool action, press ESC or .

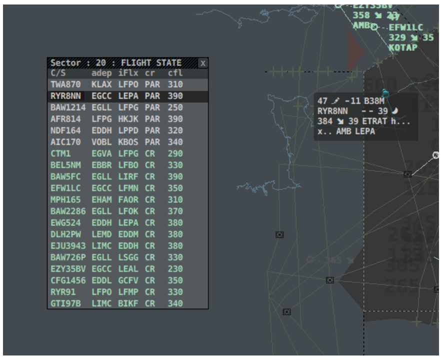
All CoFrance windows can be dragged outside of Euroscope main window and is achieved by dragging their Title bar.





Windows and lists can be drawn above flights. In order to not lose important information, if a window covers a flight concerned by an alert or a warning, the window's border color will change accordingly.

Cross highlight makes relevant flight's tag detailed when hovered from the tag itself, from a list, or from a datablock. It also details the flight's leg associated tag with .



If `Seen / Not Seen` logic is enabled, flight's information will initially appear as bold until the flight is acknowledged. If `Auto Flight leg` is enabled, cross highlighting a `Not Seen` flight will automatically display its flight leg until the flight is acknowledged or not highlighted anymore. To acknowledge a flight, on callsign.

Figure 1: Not Seen Flight.

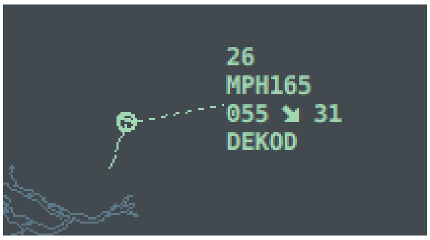
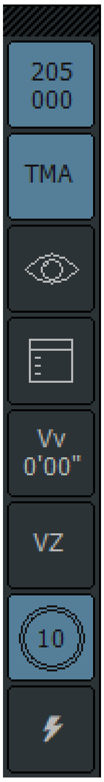


Figure 2: Seen flight after acknowledge.

@ character before callsign indicates that no flight plan has been received for this flight.

3.2 Radar Screen Overview

3.2.1 Side Menu



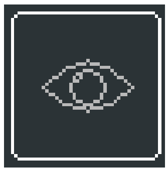
The side menu allows quick access to some radar screen parameters. The menu is movable by dragging its top bar and collapsible by it, its position is persistant across restart and stored inside `CoFrance_Preference.toml`.

This chapter details the function of each button.

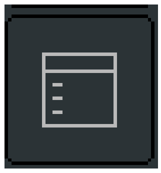
Altitude Filter
: Open the altitude filter configuration window
: Open Flight Group filter window (see 3.8)

TMA Filter
: Toggle TMA filters (see 3.8)
: Open Zone & Filters window
Blue icon indicates that TMA flights are displayed

Quick Look
: Disable all filters for 5 seconds
: Open Tag Format window (see 3.15)



Window Display
: Hide All CoFrance Window (except Side Menu)
: Open the Window Display configuration window (see 3.2.3)



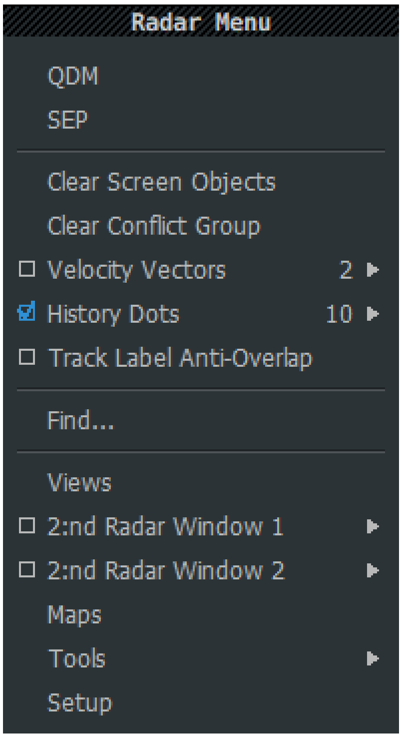
Speed Vector
: Toggle Speed vector
: Cycle Speed Vector
Scrolling on icon also allow to select length

Vertical Speed
: Toggle Vertical Speed Tagging
: –

Range Rings control
: Toggle Range rings display (see 3.7.5)
: Opens Range Rings configuration window

CPDLC
: Connect/Disconnect CPDLC (see 3.5.14)
: Toggle CPDLC aicraft list display

3.2.2 Radar Menu



The Radar menu is opened via MOD+ on the radar screen background. It offers access to several tools and menu to control radar screen image.

This chapter details the function of each button.

QDM
Activate QDM placement tool (see 3.7.2).
This action is available through StreamDeck integration.

SEP
Activate SEP placement tool (see 3.7.3).
This action is available through StreamDeck integration.

Clear Screen Objects
Clears every QDM, SEP, Route Draw from the radar screen.
This action is available through StreamDeck integration.

Clear Conflict Group
Remove every flight from the active conflict group (see 3.11).
This action is available through StreamDeck integration.

Speed Vector
Open the speed vector control sub-menu to select their length.
This action is available through StreamDeck integration.

History Dots
Open the history dots control sub-menu to select their number.

Enable Anti-Overlap
Toggle tag anti-overlap functionality (under development).

Find...
Open Find window to find VOR/FIX/NDB/CALLSIGNs (see 3.7.6).

Views
Open Views window to display predefined views (see 3.7.7).

2:nd Radar Window 1
Allow to define inset radar window 1 (see 3.7.8).

2:nd Radar Window 2
Allow to define inset radar window 2 (see 3.7.8).

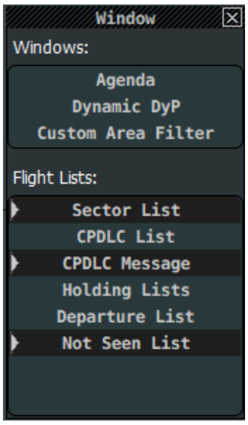
Maps
Open Maps window to toggle actively drawn maps (see 3.14).

Tools
Open Tools sub-menu to manage Range rings display (see 3.7.5).

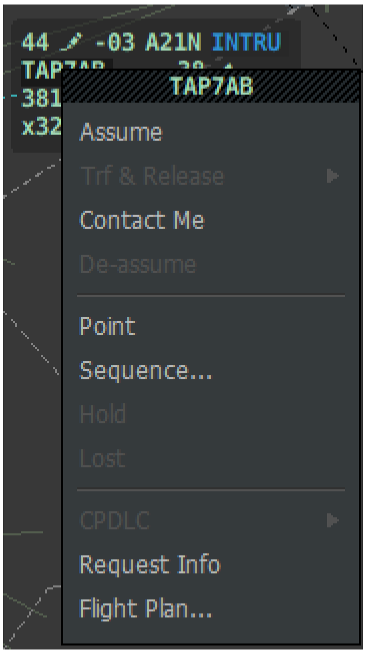
Setup
Opens the Settings window (see 2.3).

3.2.3 CoFrance Window Display Window

The CoFrance Display window allows the user to select which windows/lists are visible.
This window is opened using on `Window Display` side menu button.



3.3 Callsign Menu



The Callsign menu is opened via on the Callsign. of a flight. It offers access to several flight specific actions.

This chapter details the behavior of each function.

Assume

Your standard assume logic, used to take ownership of a previously unowned flight.

This function is only available on unowned flights.

SHRQ

Send a shoot request to the currently flight owning sector. Function compatible with `TOPSKY` sectors.

This function is only available on "owned by other" flights.

Skip

Send a coordination to the upstream sector, that you are not willing to take this flight on frequency, and delegates your exit conditions to the previous sector (see 3.6.7).

If the flight is untracked, it will just flag it as skipped (useful if the sector prediction for that flight is wrong).

Transfer Next, Trf & Release and Man Trf

`Transfer Next` is used to transfer the flight to the calculated next sector or preselected next frequency set in the extended tag (see 3.5.15). The same function is achievable by on the next sector indicator tag field.

This function is only available on assumed flights having a next sector.

`Trf & Release` opens an intermediary sub-menu to select release status of the flight. It should be the the preferred way of transferring flight ownership to another sector as `Transfer Next` will be interpreted as `not released`. Holding down ALT while selecting a release state, send hand-off clearance via CPDLC.

This function is only available on assumed flights having a next sector.

`Man Trf` opens the manual transfer menu used to transfer ownership to another sector than the calculated next sector. Two sector lists are then available `SCH (Scheduled)` displaying all the calculated sector on the flight's route and `DELEG` displaying every in-range connected sectors. Hovering an item will display the full sector indicator as well as their frequency. Clicking the sector in the list initiate the hand-off.

This function is only available on assumed flights.

Contact Me

Used to send contactme command to selected flight. A contactmeed flight will display > in front of the callsign until assumed.

This function is only available on unowned or transfer in flights.

De-assume

Used to release ownership of a flight.

This function is only allowed on assumed flights.

Point

Opens the point session popup window (see 3.6.6).

Sequence

Opens the Sequence popup window. This is used to display a number at line 0 of a flight tag's. Two sequences list available, A (PINK) and B (BLUE). Any number can be assigned to any flights (multiple flights can have the same number).
on the number to remove it.
The assigned number is local and will not be broad-casted to other online sectors.

Hold

Opens the Hold popup window (see 3.5.7). The hold assignation will be broad-casted and synchronized between all compatible sectors (LFXX).

This function is only available on assumed flights.

Lost

This function declare the flight as `LOST` and will display an alert in line0 to all compatible sectors (LFXX). This mention is used when a flight is not in contact with the current sector and will unfilter this flights for all sectors and will open the `Lost Flights` list.
To remove this state, open callsign menu and select `Cancel Lost` or `LOST` notification.

This function is only available on assumed flights.

CPDLC

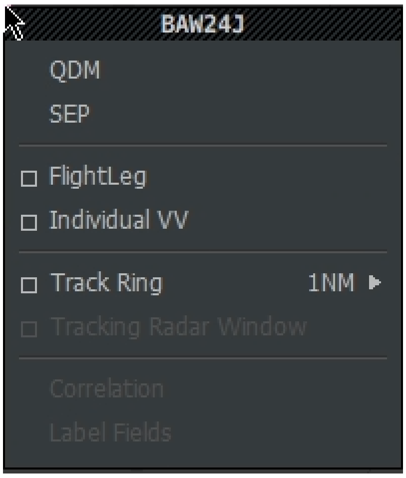
Opens the CPDLC sub-menu (see 3.5.14).

This function is only available on assumed, CPDLC-connected flights.

Request Info
Requests connection information for selected flight. Return in a private message pilot's software version and primary frequency. Can be used to quickly open a chat window with this flight.

Flight Plan...
Allow to open Flight Plan Window (see 3.5.19) or static DyP Window (see 3.7.9).

3.4 Track Menu



The Track menu is opened via MOD+ on primary radar target of a flight.

This chapter details the behavior of each function.

QDM
Activate QDM placement tool from that flight's position (see 3.7.2).

SEP
Activate SEP placement tool from that flight's position (see 3.7.3).

Flight Leg
Toggle Flight's leg display (see 3.7.1).

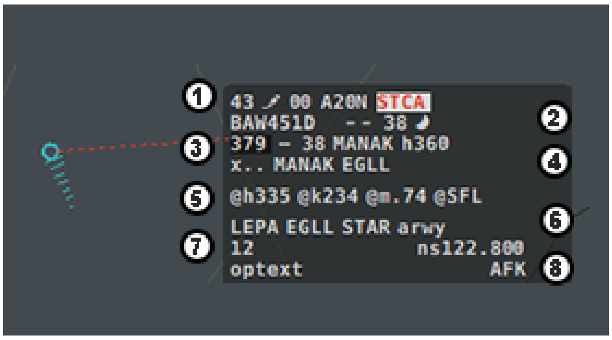
Individual VV
Toggle individual speed vector for flights. Vector's length is controlled by global speed vector setting.

Track Ring
Toggle track ring around aircraft. Size can be selected inside sub-menu.

Tracking Radar Window
Opens Tracking Radar Window in Inset Window 2 on this flight (see 3.7.8).

3.5 Interacting with Flights

3.5.1 Tag Description



Field	Description			MOD+
SPD	Current Ground Speed	Open ASPD popup	–	–
SCRATCHPAD	Scratchpad indicator	Open Scratchpad popup	Clear Scratchpad	–
VZ	Vertical Speed	–	–	–
ATYP	Aircraft Type	–	–	–
NOTIFICATION	See 3.10.4	–	Notification dependant	–

Table 1: Line 1

Field	Description			MOD+
CALLSIGN	Flight's Callsign	Open Callsign menu	–	Add to Conflict Group
SI	Next Sector Indicator	Open Man Trf popup	–	–
RFL	Requested FL	Open RFL Popup	–	–
GROUP	Conflict Group	Cycle group	Clear group	–

Table 2: Line 2

Note: for notified flights, `CFL` and `WAYPOINT` are replaced by `EFL` and `COPN` respectively.

Field	Description			MOD+
AFL	Actual FL	Draw Temp Route	Draw Perm Route	–
TENDENCY	Show Climb/Stable/Descent	–	–	–
CFL	Cleared FL	Open CFL Popup	Reset CFL	–
WAYPOINT	Cleared Waypoint	Open Waypoint Popup	Clear Direct	–
HEADING	Assigned Heading	Open Heading Popup	Clear Heading	–

Table 3: Line 3

Field	Description			MOD+
XFL	Exit FL	Open XFL Popup	–	–
COPX	Exit Waypoint	Open COPX Popup	–	–
INTENTION	Destination/Flow	Open FP Window	Open native FP Window	–

Table 4: Line 4

Field	Description			MOD+
MODESHDG	Mode S reported Heading	–	–	–
MODESSPD	Mode S reported IAS	–	–	–
MODESMACH	Mode S reported MACH	–	–	–
MODESSFL	Mode S reported FL (INOP)	–	–	–

Table 5: Line 5

Field	Description			MOD+
ADEP	Departure ICAO	–	–	–
ADES	Arrival ICAO	Open FP Window	Open native FP Window	–
STAR	Assigned STAR	Open STAR Popup	Open TRANS Popup	–
ARRRWY	Arrival Runway	Open Runway Popup	–	–

Table 6: Line 6

Field	Description			MOD+
CR	Current Owning Sector	–	–	–
NSFREQ	next Sector Frequency	Select Next Sector	–	–

Table 7: Line 7

Field	Description			MOD+
OPTEXT	FMP measures	–	–	–
AFK	Hour of return in cockpit	Open AFK Popup	–	–

Table 8: Line 8

Field	Description			MOD+
TRAJ	Trajectory Tab	Open TRAJ Tab	–	–
INFO	Information Tab	Open INFO Tab	–	–
ID	Identification Tab	Open ID Tab	–	–

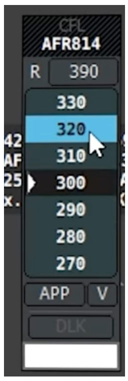
Table 9: Line 1

Line	Description			MOD+
2	Next Route Waypoints	–	–	–
3	Next Route RFL	–	–	–
4	Next Route Entry FL (INOP)	–	–	–

Line	Description			MOD+
1	Speaken Callsign	–	–	–
2	RVSM Status, 8.33kHGz Status	Open RVSM Popup	–	–
3	Aircraft in formation/Wake Turbulence	Open Formation Popup	–	–

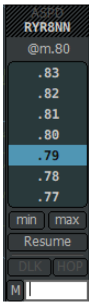
Line	Description			MOD+
1	Callsign / Current Squawk	Open Callsign Menu	–	–
2	Assigned Squawk	Open ASSR Popup	–	–

3.5.2 Assigning Level



Assigning a cleared level to a flight is done through the CFL popup window, accessed by on the CFL tag field. If a cleared level clearance was previously issued, the popup will open centered on this cleared level. For a first clearance, the popup opens on the RFL of the flight. Selecting a level in the list is done by a on said level and will close the popup.
A level can be preselected by a , closing the popup without assigning the clearance, but will allow the popup to open centered on that level for further openings. A selected level is depicted by > in front of the level.
While hovering the levels in the list, holding down MOD key enables simulated contextual filters (see 3.8).

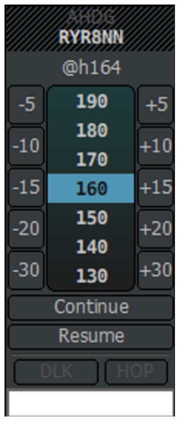
3.5.3 Assigning Speed



Assigning a speed clearance to a flight is done through the ASPD popup window, accessed by on the `SPD` or `ASPD` tag fields. The popup selects unit based on actual flight level. The min/max button are used to precise the clearance and are compatible with `TOPSKY` sectors.
Selecting `HOP` button allows to request coordinated speed to current owning sector (LFXX only).
on `ASPD` tag field, clears assigned speed.

3.5.4 Assigning Heading

Assigning a heading to a flight is done through the AHDG popup window, accessed by on the `AHDG` tag field. The window allows to clear, continue present heading, assign absolute or relative heading.
A heading can also be assigned by dragging the heading field from the tag to the desired location. On release of , the heading will be assigned. To cancel, while still dragging.
Selecting `HOP` button allows to request coordinated Heading to current owning sector (LFXX only).
on `AHDG` tag field, clears assigned heading.

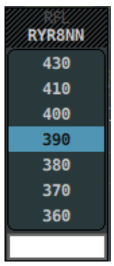


3.5.5 Assigning Direct to

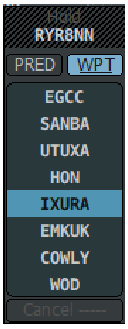
Assigning a direct to instruction is done through the WAYPOINT popup window, accessed by on the `WAYPOINT` tag field. On hover, the route is drawn and preview the direct input.
When applicable, the copx button (top most button) is enabled and displays Coordinated Exit Point.
In case you need to assign an off-route waypoint, use the input box at the bottom of the popup.
To cancel direct, simply select first waypoint of the list.

3.5.6 Modifying Final Level

Modifying final level of a flight is done through the RFL popup window, accessed by on the `RFL` tag field. The popup opens centered on current final level.



3.5.7 Assigning Hold



Assigning an holding clearance is done through the hold popup window, accessed by the `Hold` button inside the callsign menu. `Pred (Predefined)` lists published holding pattern for current sector, while `WPT (Waypoints)` lists all the flight's route waypoints. Once assigned, an indicator is shown line0 of the flight's tag.
Initially blue (unconfirmed), on it turns it white (confirmed).
Clearing a holding clearance is done through the `Cancel` button at the bottom of the window. If an EAT had been set, the holding indicator will turn yellow and remain shown until the EAT is manually cleared.

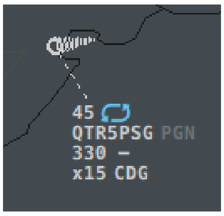


Figure 3: Hold Indicator Unconfirmed

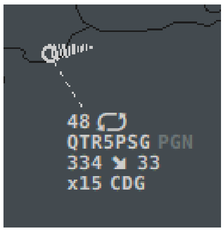
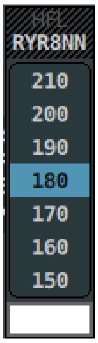


Figure 4: Hold Indicator Confirmed

3.5.8 Assigning EAT

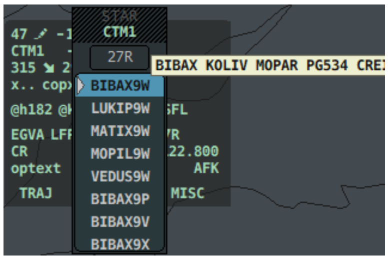
Assigning an estimated approach time is possible when a holding clearance has been issued, it is done through EAT popup window, accessed by on the `EAT` tag field.
Initially blue (unconfirmed), on it turns it white (confirmed).
The EAT is cleared by clicking `Cancel` button at the bottom of the popup.

3.5.9 Assigning HFL



Assigning a holding flight level is possible when a holding clearance has been issued, it is done through HFL popup window, accessed by on the `HFL` tag field. Once assigned, the level is shown on tag if different than `CFL`. If `CFL` is smaller than `HFL`, it will turn Yellow.
`HFL` is cleared by on the field.
on a level is used to reserve it inside the stack. It then becomes grey, and is not assignable to any flight until unreserved using again.
When assigning a predefined hold, only protected flight levels of the hold will be assignable by default.

3.5.10 Assigning SID, STAR and TRANS



Assigning SID is achieved by on SID tag field. Assigning STAR is achieved by on STAR tag field, or on TRANS tag field. Assigning TRANS is achieved by on TRANS tag field, or on STAR tag field.
The layout for those action is similar, first runway button allows to open runway assignment popup.
All procedures for this runway are then listed, hovering it will display the first 5 waypoint of this procedure. For TRANS, only transition compatible with selected runway and STAR are available.

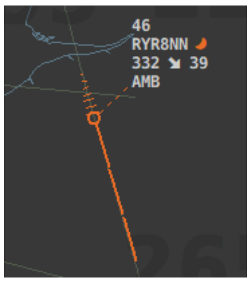
3.5.11 Assigning Departure/Arrival Runway

Assigning Departure/Arrival runway is achieved by on DEPRWY/ARRRWY tag field or using runway button from SID/STAR/TRANS assignment popup.
Runway with a green square are runway configured for this airport.
Arrival Runway can be coordinated when aircraft on *Owned by others* flights.
To propose a runway change, open RUNWAY popup and use on runway item to send coordination request to current owning sector or to force runway change (LFXX only).

3.5.12 Editing scratchpad

Editing the scratchpad string is done through the SCRATCHPAD popup window, accessed by clicking the scratchpad tag field (Pen). The list allow selection of predefined text input, while the input box allows direct keyboard entry.
on the input directly sets the scratchpad to selected string.
on the input populates the input box with the selected string and allows further entry.
Clearing the scratchpad string is done through the Clear button at the bottom of the popup.
The scratchpad string can be seen in a tool tip by hovering the SCRATCHPAD tag field.

3.5.13 Changing Aircraft Group



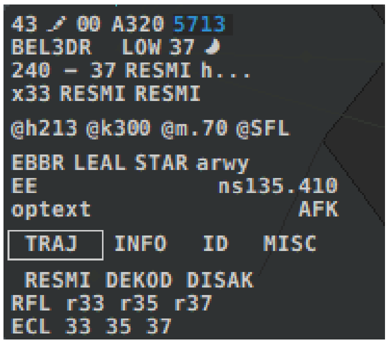
Aircraft groups are used to quickly identify tracks that require greater attention.
Aircraft group can be cycled by on the aircraft primary track or on the group tag field (moon).
Clearing aircraft group is achieved by on primary track or GROUP tag field.

3.5.14 CPDLC

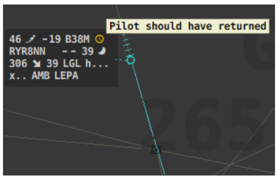
CPDLC connected aircraft are represented with a lightning strike symbol next to their callsign.
CPDLC downlinks appears below the flight's tag in blue. If multiple downlinks are received, a number will indicated the number of messages on the right of the tag field, on that number displays all the messages.
To answer that request, on the message. CPDLC uplinks are possible for `CFL`, `WAYPOINT`, `SQUAWK`, `TRANSFER` and `SPEED` clearances.
To send a clearance via CPDLC, hold ALT while opening corresponding popup or select `DLK` button inside popup.
To perform a quick CPDLC handoff, hold ALT and on `SI` tag field.
When sent, a white rectangle is drawn around the corresponding field. The color then changes based on uplink status: BLACK depicts a WILCOed uplink, YELLOW indicates an uplink in standby, ORANGE to inform that a request has been refused and RED to state that the uplink was corrupted. The rectangle disappears after a short time or by .
`Callsign Menu` CPDLC sub-menu is used to `Logoff` aircrafts, send `Assigned Squawk`, send `Stuck Mic` or send `Squawk Ident` requests.

3.5.15 Extended Tag

Extended tag allows to retrieve information about a flight that is not displayed through detailed tag. It is accessed by MOD+ or over the detailed tag of the flight. Further scrolls will open different tabs to display more information.
First extended tag line shows mode S received data.
Second line will display flight route data such as `ADEP`, `ADES`, `STAR` and arrival runway. on these will open corresponding menus.
On the next line, `CR (Comm Resp)` display the current flight owning sector, while `nsXXX.XXX` display the flight next frequency. If the flight is assumed by user, on this field allows to preselect next sector.
`OPTEXT` display FMP related information if applicable.
`AFK` is used to set AFK time.
`TRAJ` tab will display information about the next 3 waypoints of the flight's route such as their name, the final level and the entry coordinated level.
`INFO` tab display information about the flight's equipment such as RVSM & 8.33kHz capabilities, company callsign, number of aircraft in formation and wake turbulence category.
`ID` tab is all squawk related information and allows the assignation of a new code.
`MISC` tab is currently not available.



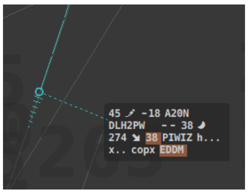
3.5.16 AFK



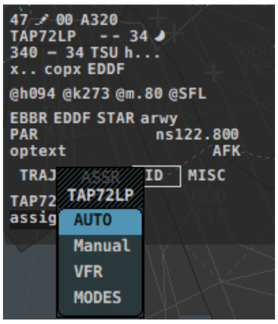
AFK Tag field is used when a pilot request AFK time. on the tag field opens the AFK popup window. Enter the UTC time when the pilot should come back or the amount of time the pilot is AFK for. The field will alway display the time at which the pilot should be back. If the field is not cleared (or `Cancel AFK`) before the time is reached, a warning is displayed line0 of the tag with a clock icon. When a flight is AFK, the mention *AFK* is displayed line0 of base tag.

3.5.17 Field Highlight

Field highlight allows to keep track of a field and might be used to specify out of ordinary or temporary clearance. It is toggled by over any tag field. The highlight is local to the controller, it will not be broad-casted to other sectors.
Highlighting a field prevents the filtering of the tag and forces it's display.



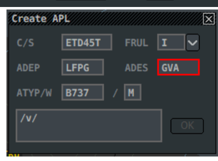
3.5.18 Assigning Squawk



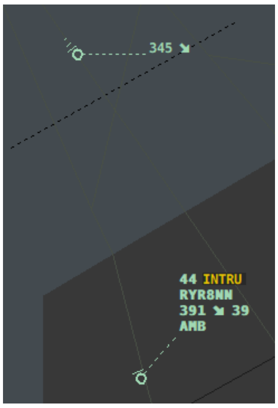
Assigning squawk is achieved throught the `ID` tab of the extended tag. on `Assigned XXXX` opens the CCAMS menu.
If the assigned squawk is different than the current entered squawk, a notification with the new code is sent to line0.
To send a squawk code via CPDLC, either use `CPDLC` sub-menu of the callsign menu and select `Set Squawk XXXX` or ALT+ `Assigned Squawk` tag field.
on ASSR notification opens the popup.

3.5.19 Flight Plan & Abbreviated Flight Plan Window

Flight Plan window displays all the important information about a flight's submitted flight plan and give the user the ability to modify them. The window is opened by on `INTENTION` or `ADES` Tag fields. When modifying the flight plan, if the rectangle around the entry becomes Red, the entry is invalid and the changes won't be able to be submitted until resolved. `Refresh` button repopulate all field using current flight plan, while `Ok` amends it.
The Flight Plan window can be closed by on close icon or by pressing ESC.
The Abbreviated Flight Plan window is a minimal Flight Plan window used to quickly create a flight plan for flight that do not have one.
To open the APL Window, open Track Menu of the flight and select `Create APL...` action.



3.5.20 Tag Minimisation



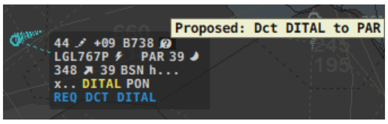
Minimizing tag allows reducing tag layout to minimum required fields to unclutter the radar display.
To be minimized, a flight should meet the following requirements:

- The flight state should not be `ASSUMED`, `TRANSFERRED OUT` or `TRANSFERRED IN`
- The flight should not have an active notification
- The flight should not be in an alert state

To (un)minimize a tag, double on the detailed tag background.
When a flight is handed-off on advisory, it's tag is automatically minimized.

3.6 Coordination

3.6.1 COPN/COPX



Entry and Exit coordinated point are automatically displayed based on flight state.
For notified flights, `COPN` is displayed instead of `NAV` in line2.
Hovering `COPN` displays assigned `NAV`.
To request a coordination, on `COPN/COPX` tag fields.
In the popup, selecting `NEGO` sends a request to tracking controller. Not selecting `NEGO` forces new entry/exit nav.
When you request a coordination, the coordinated point will be displayed in base tag and in Yellow until the coordination has been accepted, refused or has timed out.
When a coordination has been requested to you, the coordinated point will be displayed in base tag, in Pink and with a `?` character in line0.
To accept incoming coordination, on `?` notification.
To refuse incoming coordination, on `?` notification.
An accepted point coordination automatically updates the `Direct To` field.

3.6.2 EFL/XFL

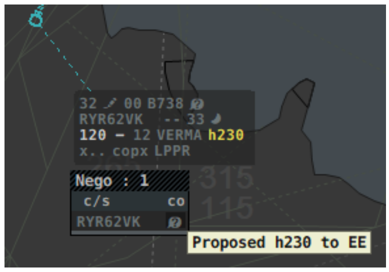
Entry and Exit coordinated altitudes are automatically displayed based on flight state.
For notified flights, `EFL` is displayed instead of `CFL` in line2.
Hovering `EFL` displays `CFL`.
To request a coordination, on `EFL/XFL` tag fields.
EFL/XFL coordination follows same color code and logic as COPX/COPN.

3.6.3 Speed Hand-Off Proposition

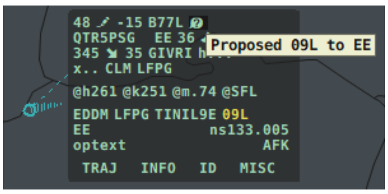
Selecting `HOP` button in `ASPD` Popup allows to request coordinated speed to current owning sector (LFXX only).
This button is only enabled on *Owned by others* flights.
The requested coordination will then show `Yellow` until accepted or refused.
As receiving sector, to accept such coordination, on `Coordination` flag or proposed ASPD and use the opened ASPD popup to assign it.
To refuse, on `Coordination` flag or porposed ASPD.

3.6.4 Heading Hand-Off Proposition

Selecting `HOP` button in `AHDG` Popup allows to request coordinated heading to current owning sector (LFXX only).
This button is only enabled on *Owned by others* flights.
The requested coordination will then show `Yellow` until accepted or refused.
As receiving sector, to accept such coordination, on `Coordination` flag or proposed AHDG and use the opened AHDG popup to assign it.
To refuse, on `Coordination` flag or porposed AHDG.



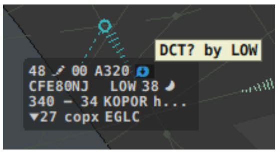
3.6.5 Arrival Runway Coordination



Arrival Runway can be coordinated on *Owned by others* flights.
To propose a runway change, open `RUNWAY` popup and use on runway item to send coordination request to current owning sector or to force runway change (LFXX only).

3.6.6 Point System

The point system allows ATC to point a flight to another compatible sector (LFXX). This action will un-filter the flight on both the sending and receiving radar screen as well as display the point message by hovering the tag point icon.
When opening the popup, next sector is automatically preselected for assumed flights and current sector for notified flights.



3.6.7 Skip

Initiating a skip. When the previous sector isn't an internal sector (LFXX), it will flag the flight as skipped for you only, and initiate a COPX and XFL coordination to delegate your exit conditions to the previous sector, as well as displaying a manual coordination flag meaning this has to be coordinated another way. However, when that sector can handle Skip coordination, it will not only initiate the COPX and XFL coordination, but also add an additional "SKIP" flag. Once the previous sector is handing over the flight, the tag will still transit by your sector and then be automatically transferred to the next one (or released if there isn't a next sector).

Receiving a skip. A received Skip, has to be either accepted or refused on the corresponding tag field. Accepting the Skip, will update the frequency of the next sector but **NOT** the actual next sector. In fact, that tag has to be handed over to the actual next sector (i.e. the one requesting a Skip). Once that sector receives the flight, it will automatically transfer its tag to the next sector. As a reminder, the next sector field will include ">" symbol.

NOTE: To cancel a skip or to refuse a skip, a manual coordination is always required.

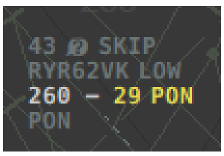


Figure 5: Initiating a skip (EE skipped)

Figure 6: Receiving a skip (EE skipped)

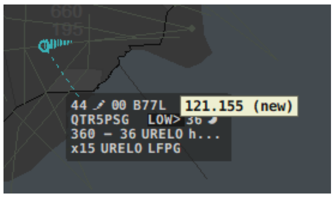
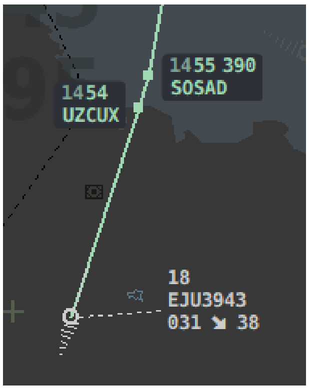


Figure 7: New frequency when skip is accepted (LOW skipped)

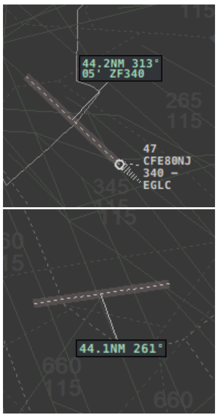
3.7 Tools

3.7.1 Flight Leg

Flight Leg displays the flight's route on radar screen along with the waypoints names, time estimates, flight status and RFL changes. To display a flight leg, use `Track Menu` button, or open `waypoint` popup or on `AFL` tag field to display temporary (hides when not hovering tag) or on `AFL` to display statically.
along a flight leg details the appropriate flight tag.
along a flight leg deletes it.



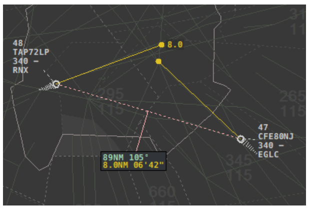
3.7.2 QDM



The QDM tool allows to get bearing and distance information between two positions. When QDM tool is enabled, the cursor turns into a crosshair, and the user needs to 2 positions, it can be 2 flights, 1 flight & a position or 2 positions. If only one flight is selected, the QDM window will display the distance, bearing, time to target and estimated altitude on target. If 2 flights are selected, the window will display distance, bearing from first to second flight, bearing from second to first as well as `SEP (separation)` info if closer than 20 nm and `CPA (closest point of approach)` if closer than 10nm.
When placed using one Flight and one position, hovering the info box shows extended informations about this QDM line.
To remove one QDM, on the QDM line.
Quick QDM activation is performed by double MOD+ on the radar background.
QDM is also available through conflict groups (see 3.11).
This action is available through StreamDeck integration.

3.7.3 SEP

The SEP tool allows to display the closest encounter between two flights based on their current track and speed. The tool will display the position of the encounter and the distance at the encounter. The color of the SEP tool is distance dependant, if the encounter is closer than 6nm, the SEP tool is RED, if closer than 9nm, it is YELLOW, otherwise it is GREEN. The maximum look-ahead time is 20 min, after which the SEP tool won't be shown. SEP is also available through conflict groups (see 3.11).
This action is available through StreamDeck integration.

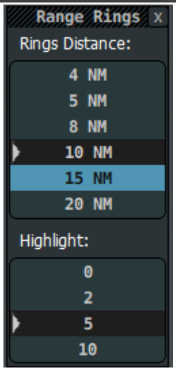


3.7.4 Route Extrapolation

Route extrapolation allows to display predicted positions of every aircraft in the conflict group. The extrapolation time is selected by dragging from a primary track of a flight in the group. Calculation is based on current route and speed.

3.7.5 Range Rings

Range rings can be displayed on the radar using the Range Rings tool. To access this tool, open the radar menu using MOD+ on the radar screen, then open `Tools` sub-menu and select `Add Range Ring`. The cursor turn into a crosshair indicating the correct activation of the tool. Use to select the rings center location, then move the cursor around to select the total range of the rings, the preview should now show on the radar, once satisfied with the distance, to confirm it. Finally, the user is able to select the angle of the rings to display partial range rings, for this, drag the cursor around to see the previewed angle until satisfied, at which point the user can to confirm the rings placement. During the angle selection, pressing MOD selects the complementary angle.
This action is available through StreamDeck integration.
To remove a range rings, select `Remove Range Rings` from the `Tools` sub-menu, the cursor turn into a crosshair indicating the correct activation of the tool. Then, the center of the rings you want to remove.
To configure the range rings, open the Range Rings Configuration Window. It allows you to select the distance between each ring and the interval between ring highlights.
Rings can also be placed using `.ring <position>` command where position is the name of an ICAO, FIX, VOR, NBD, CALLSIGN.
This action is available through StreamDeck integration.

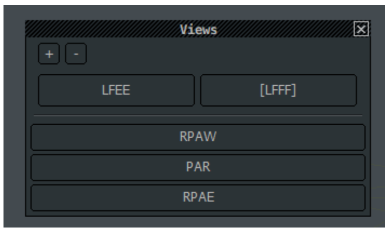


3.7.6 Find

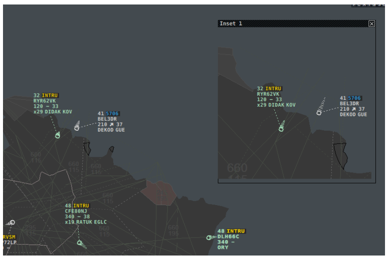
The find window is used to quickly localize requested position on the radar screen.
Via the dropdown list, you can select the type of the object you are trying to find (VOR/NDB/FIX/CALLSIGN). Then populate the text entry on the right of the window.
If the object is found, the window is close, and a line from the center of the radar screen to the requested position is drawn for 5 seconds. If it is not found, the window stays open allowing the user to correct the entry or close manually the window via the top-right close button.
This action is available through StreamDeck integration.

3.7.7 Views

The Views Window allows to Zoom and UnZoom the main radar window (similar to F11/F12 Euroscope shortcuts) but also defines preconfigured views for specific sectors to quickly adjust radar screen viewing area.
load the selected views.
save the current viewing area to selected view.
Views are defined inside `views.json` file.

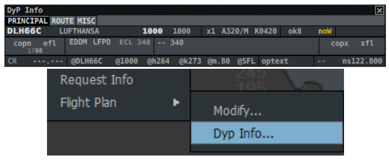


3.7.8 Inset Radar Window



Inset Radar Windows are a 1:1 recreation of the radar from which they are opened. They allow the user to isolate an area from the main radar screen.
Recall is used to reopen a previously defined inset that has been closed.
Insets can be resized, zoom and panned, independently from the main radar screen.
Inset can be opened from a flight's `Track Menu`, the inset window will then remain centered on this flight, until the inset view is panned.
Inset opened from `Track Menu` are always using Inset Radar Window 2.

3.7.9 DyP Window



Dynamic Presentation Window is another representation of all extended tag information of a flight.
This Window exists in two format:
Dynamic: opened via the `CoFrance Window Display` window and displays information of the currently selected flight.
Static: opened via the `Callsign Menu` and displays information of the currently flight used to open it. 5 Static DyP window can be opened at the same time.

3.8 Filters

Filters are used to hide flight that do not meet specific rules.
By default, all flights are filtered except:

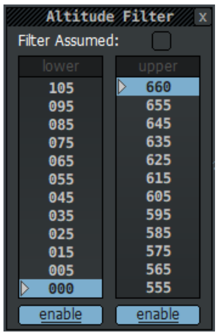
- Assumed Flights
- Flights in transfer from/to current sector
- Flights concerned by Alerts
- Flights concerned by Notification
- Flight having active coordination
- Flight having active tool
- Flight in detailed state
- Flight having field or group highlight
- Flight having special SSR codes (7400/7500/7600/7700/5677)

The setting `Filter Only Unconcerned` is used to prevent filtering of Notified flights (Green tag).
The `Quick Look` function clears all filter for 5 seconds and is accessed in the side menu.
Filters are radar screen specific, meaning that each open radar view can have different filter configuration.
Filters are saved inside `.asr` file on close, and automatically selected when reopening said `.asr` file.

This action is available through StreamDeck integration.

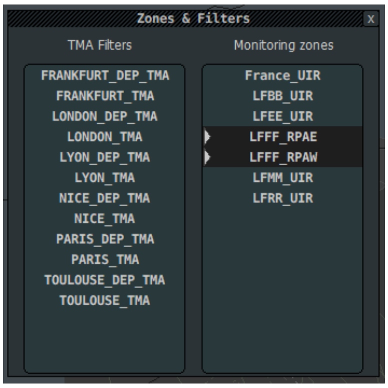
3.8.1 Altitude filters

Altitude filter controls are accessed by on the side menu icon. The lists are used to select the lower & upper levels while the `Enable` buttons are used to activate the level.
The `Filter Assumed` button is used to allow filtering of Assumed state flights. This option is mainly to allow Enroute position to have clean approach radar screen without Upper levels flights clutter the radar image. This setting should be used with caution as important information can be filtered out and thus this settings is not saved when closing `.asr` file.
The option labels turns Red when enabled.



This action is available through StreamDeck integration.

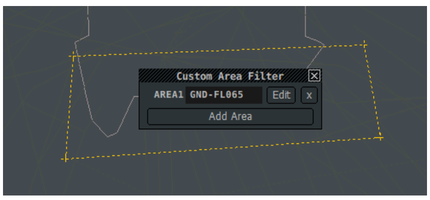
3.8.2 Geographic filters



TMA filters uses defined airspace and filter flight inside lateral and vertical limit of this area. They are usually used to filter out highly congested airspaces such as TMA when they are not of concern for the user's sector.
TMA Filters are composed of two layers, [SFC-FLOORALT] and [FLOORALT-UPPERALT]. In the first layer, all traffic is filtered whereas in the second, traffic is filtered based on user setting.
Geographic filters menu is accessed by on `TMA side-menu` button.
Monitoring zones are volumes configured as extended from the active sector (currently not shown), which defines an area of interest. Unconcerned flights within that volume are displayed with a lighter grey color indicating they should be monitored. They are processed by the plugin as different from actual unconcerned flights, interacting differently with filters for instance. Alerts such as STCA, are refreshing more often for these flights, and these alerts are shown with a higher level of urgency.
When connecting to the network, with an ACC callsign, these zones are automatically enabled.

3.8.3 Custom Area Filters

Custom Area Filters are user defined area in which the tag of an aircraft is filtered if within the flight level range.
To manage areas, open the `Custom Area Filter` window, and the `Add Area` button. Then on radar background places coordinates.
To validate the area, on the first coordinates.
To cancel the area, anywhere on the screen.
The `Custom Area Filter` window lists all the defined areas, and allows the user to modify the flight level range of each area.
You can edit an already placed area by on `Edit` button.
To delete an area, the `X` button.
Custom Area Filters are always drawn on radar background to remind the user of their presence.
These areas are not persistent across restart.



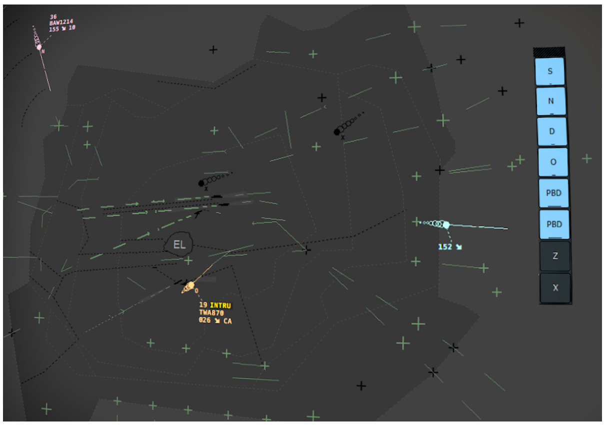
3.8.4 Contextual filters

Contextual filters are filters based on selected fields when activated. They are used to quickly clean the radar image of flight that do not meet the contextual filter's conditions. This filter is the only filter unaffected by overrides and therefore cannot be permanently activated. A warning message is displayed at the top of the screen when enabled.
This filter is activated by holding down MOD key when hovering a compatible tag field.
Compatible tag fields are:

- `INTENTION`
- `CR`
- `XFL`
- `CFL`

When the filter is used on `INTENTION`, `CR` or `XFL`, only the flights having the corresponding field equal to the selected one will remain displayed on the radar.
When the filter is used on `CFL`, only the flight inside the range [`AFL`-`CFL`] or evolving inside this range will remain displayed. The `CFL` contextual filter can be enabled in the CFL popup window to simulate CFL clearances and check radar image.

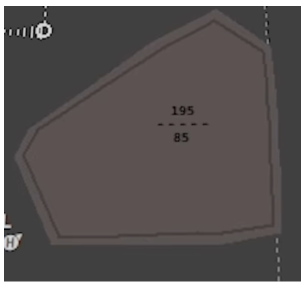
3.8.5 Flight Group filters



Flight Group filters are filters that only hide Tag of flights of a certain group.
Flight group are defined inside `tag_formats.json` file and thus, theses filters are only accessible when a tag format is being used (see 3.15).
To open the flight group filter window, Altitude filter side-menu button.
The window then displayed all available groups. Filtered groups are saved inside `.asr` file, and are automatically selected when reopening said file.

3.9 Restricted Areas

Restricted areas are dynamically displayed on the radar through the `RAMAS API`. When within 30 minutes of its activation period, an area will be displayed on the radar with the pre-active color (Grey). When the area becomes active, it changes to the active color (Red).
The lower & upper bounds of the area are alway shown in it's center, hovering them with the mouse displays the time window of the activation.



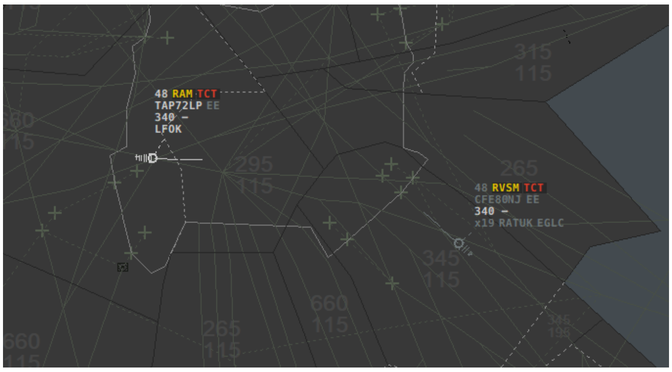
3.10 Alerts

3.10.1 STCA

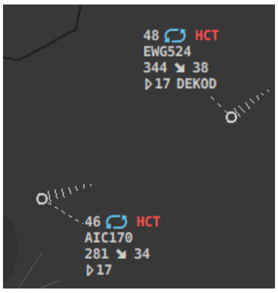
Short Term Conflict Avoidance is an alert triggered when two aircrafts are about to break minimum separation criterias. This alert is computed based on the current altitude, speed, heading and vertical speed and is therefore precise only when both aircrafts paths are not subjects to changes.
This alert does not take into account any controller assigned clearance and thus can create false positives when clearances are safe.

3.10.2 TCT

The TCT alerts on an upcoming loss of separation. There is 2 levels to this alert, depicting its urgency: Static alert indicates that the loss of separation is later than 3 minutes, while blinking alert indicates that the loss of separation will happen in the next 3 minutes. The TCT alert does take into account the flight's route or heading if assigned as well as cleared flight levels.
If no action is taken, a static TCT will become a blinking TCT.



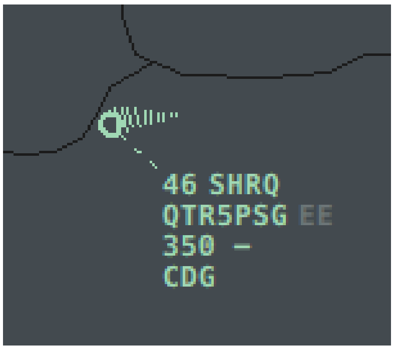
3.10.3 HCT



HCT alert is triggered when two or more aircraft are assigned the same HFL in the same holding pattern. It is depicted by the mention *HCT* in red on line0.
The alert is triggered even if flights are still not in the holding pattern but are assigned to it, allowing greater time for action.

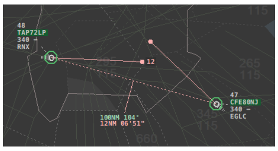
3.10.4 Notifications

Notifications are all the other information presented in line0 of the flight's tag. They can be close by .



Notification	Meaning	
EMRG	Squawking 7700	–
RDOF	Squawking 7600	–
HJCK	Squawking 7500	–
SURT	Squawking 7400	–
ATTN	Squawking 5677	–
IDENT	Squawking IDENT	–
CLAM	Cleared Level Deviation	–
RAM	Cleared Route Deviation	–
LOST	Lost radar contact	Remove Lost
AFK	Away From Keyboard	Clear AFK
FMP	Concerned by MIT	Acknoledge measure
APW	Restricted Area Proximity Warning	Clear notification
RVSM	Not RVSM capable inside/cleared into RVSM airspace	Clear notification
5705	Invalid Squawk	Clear notification
INTRU	Uncontrolled aircraft inside AOR	Clear notification
INTRU	Uncontrolled aircraft inside AOR soon	Clear notification
EFL	Non-standard entry-FL	Clear notification
OCL	Late Oceanic Clearance Request	Clear notification
F350	Oceanic level deviation	Clear notification
SHRQ	Shoot Requested	Clear notification
REL	Released Hand-off	Clear notification
STAY	Flight in STAY	Clear notification
SKIP	Skipped flight	Clear notification

3.11 Conflict Groups



Conflict groups are used to activate temporary conflict detection tools on a group of aircraft.
To add an aircraft to the conflict group, the primary radar target or MOD+ flight's callsign. When in the conflict group, the primary track will be surrounded by a green hexagon and the callsign will have a green background.
Hovering the radar target display the temporary SEP/QDM tool with another flight of the conflict group. To convert a temporary SEP/QDM to static one, press `CAPS` key while hovering the radar target.
Cycling between flights is achieved by the primary radar target.
To clear the conflict group, select `Clear Conflict Group` in the radar menu or any primary track of the group.
Saving and recalling conflict groups is done through the Agenda (see 3.12).
This action is available through StreamDeck integration.

3.12 Agenda

The agenda window allows to store conflict groups to an associated time, allowing better prioritization and an easy way to recall conflict groups on radar.
The Agenda is opened using `CoFrance Window Display Window` (see 3.2.3).
To add a conflict group to the agenda, select the concerned aircraft inside the same conflict group and either inside Agenda window at the corresponding time, or extrapolate their route to required time and maintaining MOD while releasing extrapolation.
To add a single aircraft to a saved conflict group, select the aircraft in the active conflict group and press MOD+ on the target conflict group inside Agenda window.
Hovering saved conflict group display further actions such as extrapolate to drag saved group along time line, Recall button to make this group the active conflict group or to clear the active conflict group, tag grouping buttons to change color, as well as groupe delete button and aircraft delete buttons.
When an aircraft is in a datablock, a chain like symbol is displayed left of its callsign.
This action is available through StreamDeck integration.

3.13 Lists

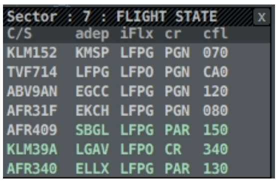
`CoFrance` uses 2 types of lists: *openable lists* and *popup lists*.
Openable list are displayed through radar menu `Flight Lists` button and can be closed by the close icon.
Popup lists are automatically displayed when not empty and are not closable manually.

List Title shows number of flights in list as well as what ordering logic is currently being used.

By Default, a list is ordered by Flight State (ASSUMED > TRANSFERRED > NOTIFIED > UNCONCERNED).

MOD+ on list title bar opens list menu to manage its content & settings.

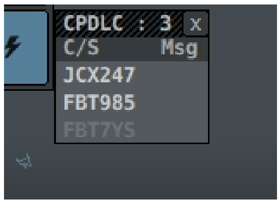
3.13.1 Sector List



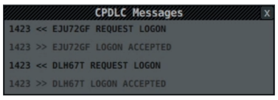
Sector list displays flights that are `Assumed`, `Transferred In`, `Transferred Out` or `Notified`.
Color in the list reflect flight's state.

3.13.2 CPDLC List

CPDLC List display actively CPDLC connected flights along with down-link request indicator (M) if present.
on `Message Received` indicator allows the user to answer the request.



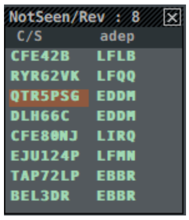
3.13.3 CPDLC Message list



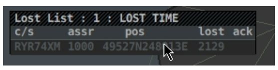
The CPDLC message list shows previous and ongoing CPDLC communications.
Chevron direction indicates Sender.
Messages are automatically deleted after 10 minutes.
double on any message deletes it immediately.

3.13.4 Not Seen List

The Not Seen List displays all notified flights within 15 minutes of sector entry that are not marked as seen by the user.



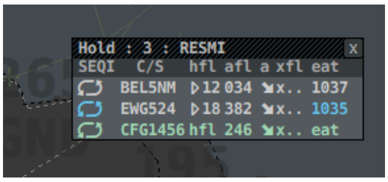
3.13.5 Lost List



Lost list display Lost aircrafts, manually by selecting `Lost` inside the callsign menu or automatically when the aircraft disconnects. When the flight disconnects, it remains displayed on the radar screen in a ghost state so as to remember where the last signal was received. To remove a ghost flight, open the callsign menu and select `Delete Flight`.
Acknowledging a flight in this list, removes it from the list.
This list is Popup.

3.13.6 Holding List

The holding list is a holding waypoint dependent list, and is automatically opened with the first aircraft assigned to a holding pattern. The list displays all the flights currently assigned to it. Each list can be independently closed, to recall a closed holding list, select `Holding List` inside `Flight List` sub-menu of the radar menu.
This list is Popup.

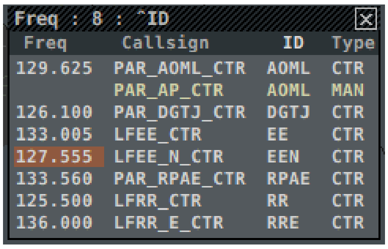


3.13.7 Negotiation List

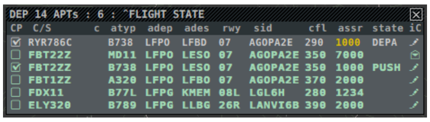
The Negotiation List is displayed when a negotiation is ongoing with at least one flight. It allows to quickly find the flights concerned by a coordination using cross highlight.
This list is Popup.

3.13.8 Frequency List

The frequency list displays online frequencies and manually added ones.
Using list menu, the user can select the displayed columns and number of entries before scrolling is required as well as add `Ghost` frequencies (Yellow).
`Ghost` frequencies are sub-sectors planned to come online soon and for which transfers are requested on appropriate frequency.
Adding a `Ghost` frequency allows the user to have the sector already visible and be able to display in flights' tag.
To add a `Ghost` Frequency, open list menu and select `Add Frequency` action. Then select the current online ATC covering the soon to be opened sector. Finally input said sector's callsign and click `Add` button.
When frequencies have been added, use on `SI` tag field to cycle between all sector frequencies of the Coordinated Next Sector.
`Ghost` frequencies are automatically removed if the sector comes online or if the associated ATC disconnect.



3.13.9 Departure List



Departure list allows see on ground flights from selected ICAOs.
When opening a Departure list, a window opens prompting the user to enter a list of comma separated ICAO codes for which to display departures.
The `Default` button is used to select every currently owned airports.
Multiple departure lists can be opened using different configurations.

3.13.10 Waypoint List

The Waypoint list allows to see all flight having said waypoint in there route.
When opening a waypoint list, user is prompted to enter a Waypoint (can be a FIX, VOR, NDB).
This list can be useful when presequencing arrivals on a waypoint.

3.14 Maps

Maps are user/system defined polygons, lines or text to be displayed dynamically on the radar screen background. They can be toggled via `Maps` button inside the radar menu.
Maps can be defined `Active` by default or based on defined conditions.
Active maps are saved in `.asr` file on close and automatically activated when reopening said `.asr` file.

3.15 Tag Formats

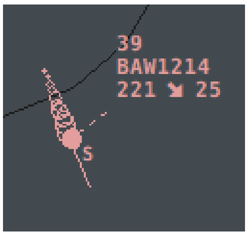
Tag format allows the user to define flight groups with custom rules to override default colors, symbols and layout, to better depicts each approach radar tag formatting.
Tag Formats are defined inside `tag_formats.json` file.
To enable a Format, open the Tag Format window using 🖱 on `Quick Look` side menu button.
This window will display all available formats.
Selected Tag Format is saved inside `.asr` file on close, and is automatically reloaded when reopening said `.asr` file.
To select a new Format, 🖱 on associated button, the radar image should immediately update.
Tag Format are `.asr` specific, meaning that if multiple radar screen are opened, each can have its own formatting.
Flights belonging to a flight group are, by default, not filterable by altitude filters. Specific groups can be defined as filterable inside `tag_formats.json` file.

The next section will explain the already available Tag Formats symbology.

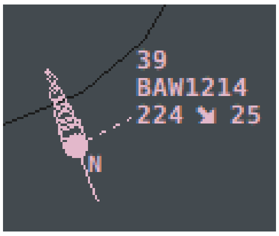
Group refers to the associated button name inside the Flight Group Filter Window.

3.15.1 CDG

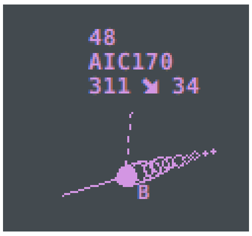
PG/PB Departure
Group: D



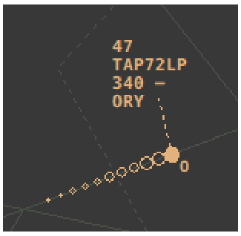
PG South Rwy Arrival
Group: RNSB



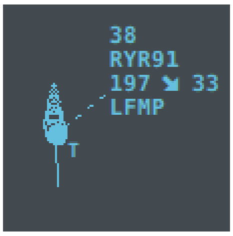
PG North Rwy Arrival
Group: RNSB



PB Arrival
Group: RNSB

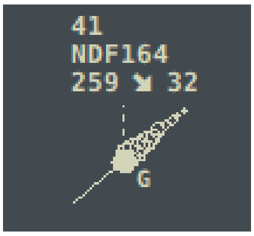


PO Arrival
Group: L

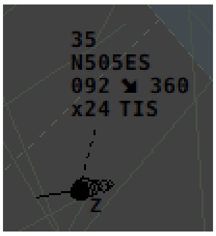


PO Departure
Group: M

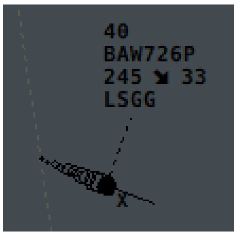
PV Arrival
Group: V



Local Flight
Group: GP

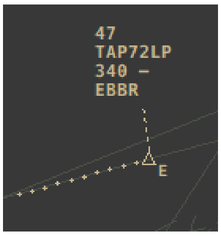


VFR Transit
Group: Z

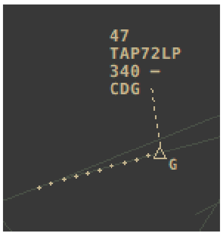


IFR Transit
Group: X

3.15.2 Orly

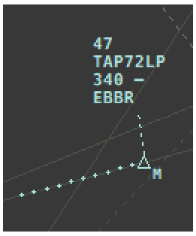


PG/PB Departure
Group: E

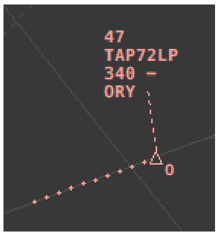


PG Arrival
Group: G

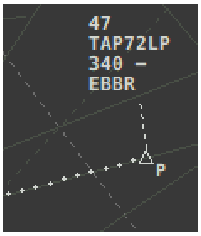
PB Arrival
Group: K



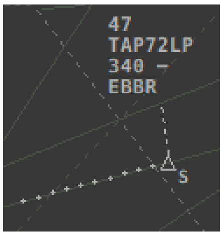
PO/PN/PV Departure
Group: M



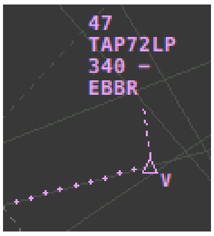
PO Arrival
Group: O



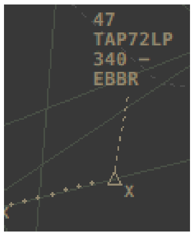
Local Flight
Group: P



VFR Transit
Group: S

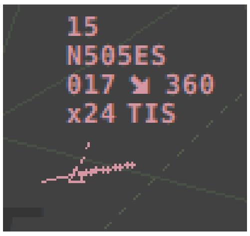


PN/PV Arrival
Group: V

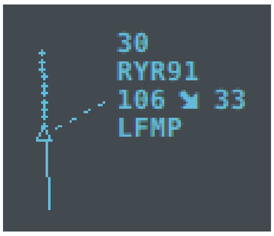


IFR Transit
Group: X

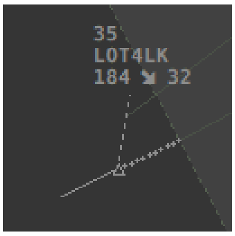
3.15.3 IRMA



Departure Flight
Group: DEP

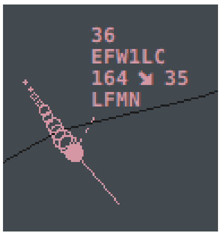


Arrival Flight
Group: ARR

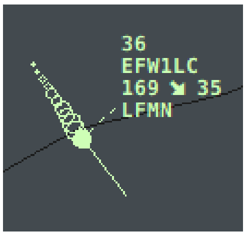


Transit Flight
Group: IFR/VFR

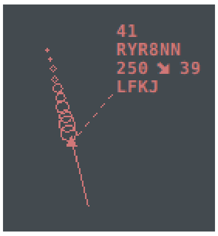
3.15.4 Nice



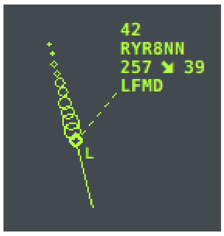
Nice Departure
Group: DEP



Nice Arrival
Group: ARR



Satellite Departure
Group: SATDEP



Cannes Arrival
Group: MDARR

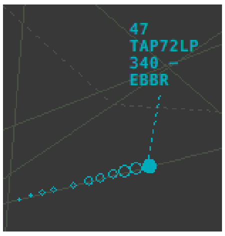
Hyères Arrival
Group: THARR

Le Luc Arrival
Group: MCARR

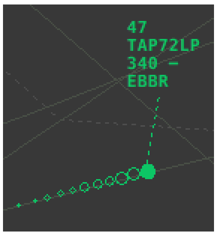
VFR Transit **Group: VFRTRS**

IFR Transit
Group: IFRTRS

3.15.5 Lyon

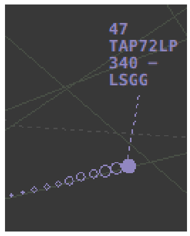


IFR Transit
Group: Tous IFR

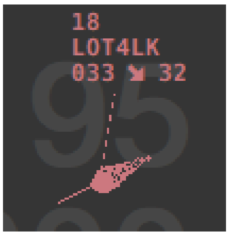


LL Arrival
Group: ARR

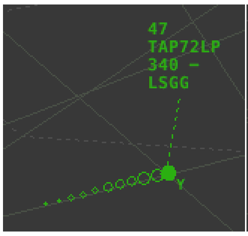
LY Departure
Group: DEP



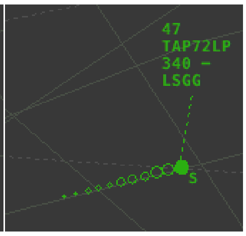
LSGG Dep/Arr
Group: Tous IFR



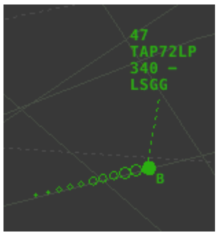
LL Departure
Group: DEP



LY Arrival
Group: ARR



LS Arrival
Group: ARR



LB Arrival
Group: ARR

IFR Transit
Group: Tous VFR